



MORADABAD INSTITUTE OF TECHNOLOGY
APPLIED SCIENCES & HUMANITIES DEPARTMENT
SESSIONAL TEST - I

Set - I

Course: B.Tech.
Session 2024-25
Subject Name: MATHS-I
Max. Marks: 20

Semester: I
Section: A, B, C, D, E, F, G, H
Subject Code: (BAS-103)
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	2	1	1	1	1	2
Bloom's Level	K3	K3	K3	K3	K3	K3

Note: (1) This paper contains three sections, Section A, Section B & Section C.
(2) All sections are compulsory.

SECTION-A		2*1=2 Marks
Q.1.	If $y = \cos^3 x$ then find y_n .	
SECTION-B		3*2=6 Marks
Q.2.	Find the inverse of the matrix by elementary Row transformation $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$.	
Q.3.	Find the rank of the matrix by reducing it to Normal form $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$	
SECTION-C		4*3=12 Marks
Q.4.	Investigate for consistency of the following equations and find the solution for x,y,z. $3x + 3y + 2z = 1,$ $x + 2y = 4,$ $10y + 3z = -2,$ $2x - 3y - z = 5$	
Q.5.	If $A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$ then compute A^{-1} using Cayley-Hamilton theorem.	
OR		
Find the Eigen values and corresponding Eigen vectors of the matrix		$\begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.
Q.6.	If $y = e^{m \cos^{-1} x}$ then show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$ and calculate $y_n(0)$.	



MORADABAD INSTITUTE OF TECHNOLOGY
APPLIED SCIENCES & HUMANITIES DEPARTMENT
SESSIONAL TEST - I

Course: B.Tech.

Session 2025-26

Subject Name: MATHS-I

Max. Marks: 20

Semester: I

Section: A, B, C, D, E, F, G, H

Subject Code: (BAS-103)

Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	1	2	1	3	1	2
Bloom's Level	K3	K3	K3	K3	K3	K3

Note: (1) This paper contains three sections. Section A, Section B & Section C.

(2) All sections are compulsory.

SECTION-A		2*1=2 Marks
Q.1.	Show that the matrix $A = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ -i & -1 \end{bmatrix}$ is unitary.	
SECTION-B		3*2=6 Marks
Q.2.	If $u = \cos^{-1} \left[\frac{x^3 + y^3 + z^3}{ax + by + cz} \right]$ then show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 2 \cot u = 0$.	
Q.3.	Find the rank of the matrix by reducing it to Normal form $A = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 4 \\ 1 & 2 & 3 & 4 \\ -1 & -2 & 6 & -7 \end{bmatrix}$	
SECTION-C		4*3=12 Marks
Q.4.	If $x + y + z = u, y + z = uv, z = uvw$, then show that $\frac{\partial(x, y, z)}{\partial(u, v, w)} = u^2 v$	
Q.5.	Investigate for consistency of the following equations and find the solution for x,y,z. $3x + 3y + 2z = 1, x + 2y = 4, 10y + 3z = -2, 2x - 3y - z = 5$	
<u>OR</u>		
	If $A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ then compute A^{-1} using Cayley-Hamilton theorem.	
Q.6.	If $y = (\sin^{-1} x)^2$ then show that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2 y_n = 0$ and calculate $y_n(0)$.	

MORADABAD INSTITUTE OF TECHNOLOGY

APPLIED SCIENCES & HUMANITIES DEPARTMENT

A1

CLASS TEST - II

Course: B.Tech.

Session 2025-26

Subject Name: MATHS-I

Max. Marks: 20

Semester: I

Section: A, B, C, D, E, F, G, H

Subject Code: (BAS-103)

Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	5	5	5	3	4	3
Bloom's Level	K3	K3	K3	K3	K3	K3

Note: (1) This paper contains three sections. Section A, Section B & Section C.

(2) All sections are compulsory.

SECTION-A		2*1=2 Marks
Q.1.	Find Curl of a vector field given by $\vec{F} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$.	
SECTION-B		3*2=6 Marks
Q.2.	Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point P (1,2,3) in the direction of the line PQ where Q is the point (5,0,4).	
Q.3.	Evaluate Green's theorem in plane for $\int_C [(x^2 + 2xy)dx + (y^2 + x^3y)dy]$ where C is the Square having vertices at the points (0,0), (1,0), (1,1), (0,1).	
	Evaluate Stokes theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken round the square bounded by the lines $x = \pm 1, y = 0, y = 1$.	
SECTION-C		4*3=12 Marks
Q.4.	Expand $e^x \cos y$ in the powers of x & y as far as terms of third degree.	
Q.5.	Show that $\int_0^{\pi} \frac{1}{1+n \cos x} dx = \frac{\pi}{\sin n\pi}$ by using $\frac{\pi}{\sin n\pi} = \int_0^{\infty} \frac{x^{n-1}}{(1+x)^2} dx$ where $0 < n < 1$. Also find the value of $\int_0^{\pi} \frac{1}{1+5 \cos x} dx$ <u>OR</u> The plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ meets the axes in A, B, C. Apply Dirichlet's Integral to find the volume of Tetrahedron OABC. Also find its mass if the density at any point is $kxyz$.	
Q.6.	The temperature T at any point (x, y, z) in space is $T = 400xyz^2$. Find the highest temperature at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$.	
	<u>OR</u>	
	Examine for maximum and minimum values for: $\sin x + \sin y + \sin(x+y)$.	



In pursuit of excellence

MORADABAD INSTITUTE OF TECHNOLOGY
APPLIED SCIENCES & HUMANITIES DEPARTMENT
SESSIONAL TEST - II

Set A

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Session 2024-25
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Max. Marks: 20

Semester: I
Section: A, B, C, D, E, F, G, H
Subject Code: (BAS-103)
Time: 1 Hour

Q. No. :	1	2	3	4	5	6
CO No. :	2	3	2	2	3	3
Bloom's Level	K3	K3	K3	K3	K3	K3

Note: (1) This paper contains three sections. Section A, Section B & Section C.
(2) All sections are compulsory.

	SECTION-A	2*1=2 Marks
Q.1.	If $y = \frac{5x+12}{x^2+5x+6}$ then find y_n .	
	SECTION-B	3*2=6 Marks
Q.2.	Verify Chain Rule for Jacobean for $x = r\cos\theta$, $y = r\sin\theta$	
Q.3.	If $y = \sin(m \sin^{-1} x)$, prove that $(1-x^2)y_{n+2} - (2n+1)x y_{n+1} - (n^2-m^2)y_n = 0$.	
	SECTION-C	4*3=12 Marks
Q.4.	If $u = \sin^{-1} \left[\frac{x^3+y^3+z^3}{ax+by+cz} \right]$ then show that $x \frac{\partial y}{\partial x} + y \frac{\partial y}{\partial y} + z \frac{\partial y}{\partial z} = 2 \tan u$	
Q.5.	Expand $e^x \cos y$ in the powers of x & y as far as terms of third degree. OR Examine for maximum and minimum values for: $\sin x + \sin y + \sin(x+y)$.	
Q.6.	If $x+y+z=u$, $y+z=uv$, $z=uvw$, then show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$ OR Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube.	